

# Junsu Kim

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## Research Interests

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Quantum Computing, Memory Systems, Computer Architecture, Systems for ML & ML for Systems

## Education

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**Pennsylvania State University, University Park, PA** Aug. 2021 - Mar. 2022, Not Completed  
Ph.D. in Computer Science and Engineering (GPA: 3.84/4.0)

**Hanyang University, Seoul, Korea** Mar. 2014 - Feb. 2021  
B.S. in Electronic Engineering (GPA: 3.81/4.0, Graduating with Honors - Summa Cum Laude)

***Relevant Courses:** Computer Architecture (Instructor: Mahmut Kandemir, PSU), Operating System (PSU), Data Structure, Artificial Intelligence, Embedded System, SoC Design, VLSI, Microelectronic Circuits, Linear Algebra*

## Publications

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[In Preparation] Yujin Kim, Minsoo Kang, **Junsu Kim**, and Suhyun Kim. “Intergrative Solution for Catastrophic Forgetting in Data-Free Class Incremental Learning”.

[C1] Kwangrae Kim, Jeonghyun Woo, **Junsu Kim**, and Ki-Seok Chung. “HammerFilter: Robust Protection and Low Hardware Overhead Method for RowHammer”. The 39th IEEE International Conference on Computer Design (ICCD), 2021

[Poster] Kwangrae Kim, **Junsu Kim**, Jeonghyun Woo, and Ki-Seok Chung. “HammerFilter: Robust Protection and Low Hardware Overhead Method for Row-Hammering”. The 58th IEEE Design Automation Conference (DAC) Work-in-Progress, 2021

## Work Experience

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**Korea Institute of Science and Technology, Seoul, Korea** Jul. 2022 - Present  
Research Assistant at Korea Data Science Team (KDST) Supervisor: Dr. Suhyun Kim

**Pennsylvania State University, University Park, PA** Aug. 2021 - Mar. 2022  
Research Assistant at Lab of Green & Secure Integrated Circuit Systems (LOGICS) Advisor : Prof. Swaroop Ghosh

**Hanyang University, Seoul, Korea** Dec. 2019 - Mar. 2020, Aug. 2020 - Nov. 2020  
Research Assistant at Embedded System on Chip Laboratory (ESOC Lab) Advisor: Prof. Ki-Seok Chung  
Research Assistant at Computer Architecture and System SW Lab (CASS Lab) Advisor: Prof. Yongjun Park

**School for the Blind, Chuncheon, Korea** Mar. 2017 - Feb. 2019  
Assistant Teacher (Alternative Military Service)

## Research Projects

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**Fast and Robust Ensembled Quantum Computing for Variational Quantum Algorithm (VQA)** Oct. 2022 - Present  
Supervisor: Dr. Suhyun Kim, Korea Institute of Science and Technology  
♦ Training VQA with quantum machine is extremely slow due to cloud-based access, and the noise degrades the performance of VQA  
♦ EQC (ISCA'22) proposed an ensemble training using multiple quantum computers to accelerate training and mitigate noise  
♦ Observed the performance of VQA trained with EQC framework varied depending on device and its current noise level  
♦ Proposed an ensembled quantum computing framework for VQA that is fast and robust regardless of device and noise change  
♦ The framework selected a primary device that was more accessible and reliable, then optimized the given circuit for the device  
♦ During training and inference, the framework compensated for the difference in noise levels between the primary device and others  
♦ Tools: Qiskit, Python, torchQuantum

**Integrative Solution for Catastrophic Forgetting in Data-Free Class Incremental Learning** May. 2022 - Present  
Supervisor: Dr. Suhyun Kim, Korea Institute of Science and Technology  
♦ Data-Free Class Incremental Learning (DFCIL) scenario enables a model to continuously learn without violating privacy  
♦ Observed previous methods and their synthetic data caused bias in classification head, which exacerbated catastrophic forgetting  
♦ Developed a weight-balancing method to correct the bias in the classification head and a hybrid knowledge distillation approach  
♦ As the third author, proposed the feature distillation scheme, produced the main results, and actively participated in paper writing  
♦ Passed the first phase of AAAI'23, submitted to ICML'23  
♦ Tools: Pytorch, Python

## Heterogeneous Learning Rates for Hybrid Quantum-Classical Networks

Advisor: Prof. Swaroop Ghosh, Pennsylvania State University

Aug. 2021 - Nov. 2021

- ◇ Classical Neural Networks (CNN) and Quantum Neural Networks (QNN) have different parameter scales (e.g., QNN has  $0-2\pi$ )
- ◇ Using the same learning rate for both classical-quantum neural networks may cause slow convergence
- ◇ Adopted different learning rates for quantum network and classical network based on the fact that parameter scales are different
- ◇ Observed that the model performance highly depends on the learning rate of the classical neural network
- ◇ Tools: PennyLane, Tensorflow, Python

## HammerFilter: Robust Protection and Low Hardware Overhead Method for RowHammer

Advisor: Prof. Ki-Seok Chung, Hanyang University

Aug. 2020 - Nov. 2020

- ◇ A recent study showed newer chips are more vulnerable to Rowhammer (i.e., Rowhammer threshold decreased from 139K to 10K)
- ◇ Prior solutions incurred significant area overhead and/or showed imperfect protection
- ◇ Proposed a robust and low overhead RowHammer protection scheme by modifying counting bloom filter
- ◇ As the third author, participated in implementing related works, experiments, and paper writing
- ◇ Tools: DRAMSim2, Gem5, C++, Shell script

## Energy-efficient Auto-refresh Skipping Weak Rows

Personal Project

Jul. 2020 - Aug. 2020

- ◇ The auto-refresh scheme guarantees all DRAM cells not to lose data with a refresh interval of 64ms
- ◇ RAIDR(ISCA'12) showed fewer than 1000 DRAM cells out of  $10^{11}$  cells required a refresh interval shorter than 256ms
- ◇ Applied the refresh rate for robust DRAM cells to all DRAM cells based on the assumption modifying OS not to use weak rows.
- ◇ Proposed method averaged 30% improvement of refresh energy over Retention time-aware Intelligent DRAM Refresh (RAIDR)
- ◇ Tools: DRAMSim2, Gem5, C++, Shell script

## Optimize Deep Learning Computation on Edge Devices via Compiler

Advisor: Prof. Yongjun Park, Hanyang University

Dec. 2019 - Mar. 2020

- ◇ Observed the low performance of TASO (SOSP'19) when using CPU
- ◇ First learned the way to think critically, got familiar with research environments (e.g., ubuntu, docker, git)

# Coursework Projects

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## Computer Architecture Labs

Pennsylvania State University

CMPEN 431 (Computer Arch)

Aug. 2021 - Dec. 2021

- ◇ The project goal was to find the best performance and the best energy-efficient architecture design using SimpleScalar
- ◇ Searching all cases with 18 different design points (e.g., cache size) for the project goals required infeasible iterations
- ◇ Implemented heuristic searching that adjusted all parameters within a loop then observed the performance and energy-efficiency
- ◇ The searching algorithm ended when the configuration results converged (i.e., if the results was the same with the previous loop)
- ◇ Observed IPC of various cache designs under 6 workloads from SPEC CPU 2006

## Operating System Labs (with C language)

Pennsylvania State University

CMPSC 473 (Operating System)

Aug. 2021 - Dec. 2021

- ◇ Modified miniOS to set up simple system call and page table (4KB, 4-level page tables for x86\_64 architecture)
- ◇ Implemented 64bit dynamic memory allocator (malloc) with best-fit approach and segregated list

## CAPTCHA Project: Building a Machine Reads Distorted Text

Hanyang University

Intro to Artificial Intelligence

May. 2019 - Jun. 2019

- ◇ Built a framework that classified distorted text with small training data (dataset from Wilhelmy, Rodrigo Rosas, Horacio)
- ◇ Built pre-processing pipeline that detected each letter and applied data augmentation to the training dataset
- ◇ Exploited transfer learning approach to overcome over-fitting
- ◇ Achieved 3rd highest score in the class Kaggle competition and the honor of the best presentation

## Implementing HDL Programs on FPGA

Hanyang University

SoC design

May. 2019 - Jun. 2019

- ◇ Implemented FIR Filter, LED, seven segments, and VGA controller on Altera DE1-SoC FPGA using Quartus

## Embedded System Software Design

Hanyang University

Embedded System Design

May. 2019 - Jun. 2019

- ◇ Cross-compiled bootloader and kernel for Achro 210T board
- ◇ Built LED controller for Achro 210T board, and modified the display system

# Skills

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C/C++, Python, Tensorflow, Pytorch, PennyLane, Qiskit, Git, Shell script, ARM assembly, Verilog